



MSc in Computational Software Techniques in Engineering

Computational Fluid Dynamics analysis of an aircraft antenna fairing

Abstract

Fuselage-mounted Ka-band antennas enable in-flight connectivity but introduce aerodynamic penalties. Simplified fuselage + fairing modeled (CATIA V5); 2D sectional RANS (Fluent, SST $k-\omega$, $y^+ \approx 1$) at cruise flight conditions .

Objectives

- Quantify aerodynamic loads and flow disturbances from Ka-Band antenna fairing
- Compare fuselage sections with/without fairing
- Capture near wall physics with a $y^+ = 1$ mesh and SST $k-\omega$ turbulence model

Methods

- Geometry: Simplified fuselage + antenna CAD (CATIA V5); 2D slices at 0–250 mm.
- Domain: Inlet/outlet + refined antenna sector.
- Meshing: Edge-biased layers; first cell from $y^+ \approx 1$.
- Solver: ANSYS Fluent (steady RANS); SST $k-\omega$; cruise at 12.5 km, $U \approx 230$ m/s; 1000 iterations.



Flight Conditions & Key Numbers

Altitude: 12.5 km $\rightarrow \rho \approx 0.33 \text{ kg}\cdot\text{m}^{-3}$, $\mu \approx 1.32 \times 10^{-5} \text{ Pa}\cdot\text{s}$.
Velocity: ≈ 230 m/s (Mach 0.75–0.8).
 $Re \approx 2.2 \times 10^8$ (fully turbulent).
Wall resolution: $y(y^+ = 1) \approx 5.6 \text{ }\mu\text{m}$.

Key Results

Section 0 (strongest): suction pocket on forward shoulder; localized pressure rise at aft fillet; compact wake and turbulence hotspot.

Sections 50–150: similar, weaker; short separation bubble (reattachment within $\sim h$).

Sections 200–250: wake recovers to $\geq 0.9 U_\infty$ by 3–4h; turbulence only $\sim 1.5\text{--}2.3\times$ baseline.

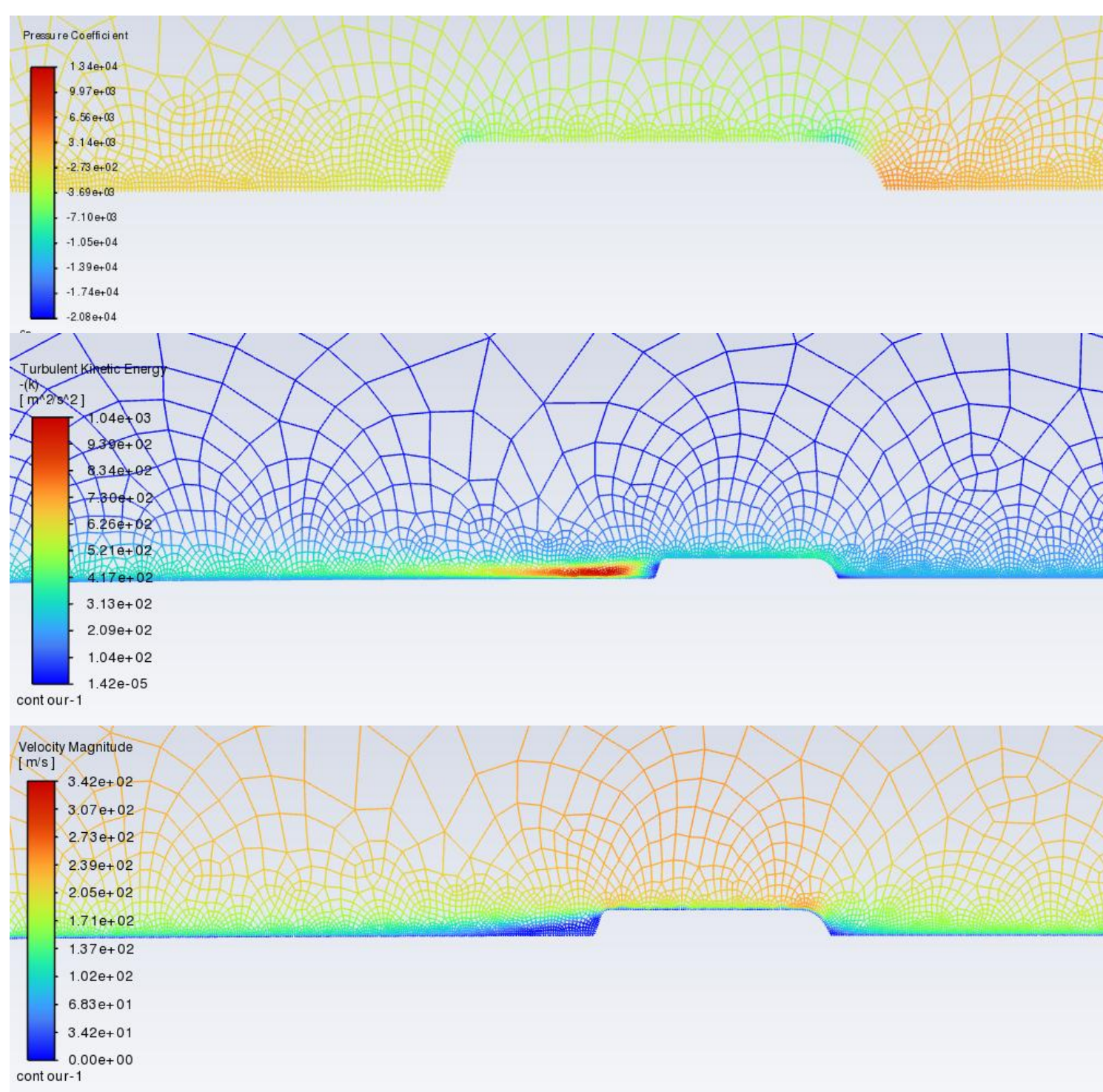
Overall: incremental drag dominated by compact form drag; downstream effects limited.

Conclusions

Fairing-induced penalties are localized; aft-fillet shaping reduces separation & drag.

2D $y^+ \approx 1$ + SST $k-\omega$ is a robust reduced-order method.

Solid foundation for future 3D and aero-structural integration.



Sylvain MAZEL

Dr. Peter Sherar

David Thornhill



Sabena Technics BGC